

Supplementary Information

Sr₂PrSbO₆ Double Perovskite as a Solar-Blind UV Photodetector: A Combined DFT and Device Simulation Study

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This supplementary document contains Supplementary Methods, three static figures (Figs. S1–S3), five animated figures (Movies S4–S8), two tables (Tables S9–S10), Supplementary Note 1 (Figs. S11–S14) presenting the thermoelectric transport properties of Sr₂PrSbO₆, Supplementary Note 2 (Table S15) presenting the mechanical properties, Supplementary Note 3 (Table S16) presenting the branch-point energy and band-edge effective masses, Supplementary Note 4 (Fig. S17, Table S18) presenting the formation enthalpy and its temperature dependence, and Table S19 listing the photodetector benchmark data of Fig. 8, followed by Supplementary References. Each animation is provided as a GIF file (for direct embedding) and an MP4 file (for video playback).

Supplementary Methods — Thermoelectric transport calculations

Lattice thermal conductivity κ_L was computed by iteratively solving the phonon Boltzmann transport equation (BTE) as implemented in phono3py^{S1–S3}. Second-order harmonic interatomic force constants (IFCs) were computed using a 2×2×1 supercell and third-order anharmonic IFCs using the 1×1×1 primitive cell, with finite atomic displacements of 0.03 Å. All single-point force calculations were performed with VASP^{S4,S5} using the projector augmented-wave (PAW) method^{S6,S7} and the generalized gradient approximation in the Perdew–Burke–Ernzerhof (PBE) parametrization^{S8}. The phonon BTE was solved on a 15×12×9 reciprocal-space mesh (1620 q-points; 476 irreducible grid points), with individual scattering-rate calculations performed independently for each irreducible grid point and merged to construct the full scattering-rate tensor. Temperatures were sampled from 100 K to 1200 K in steps of 50 K.

Electronic transport coefficients were computed within the constant relaxation-time approximation ($\tau = 10^{-14}$ s) using the BoltzTraP code^{S9}. The chemical potential μ was varied over the range $-1.5 \leq \mu - E_F \leq 0.2$ eV relative to the DFT Fermi energy to simulate both n-type and p-type doping conditions. The total figure of merit was evaluated as $ZT = S^2\sigma T/(\kappa_e + \kappa_L)$, where κ_L at each temperature was obtained by cubic-spline interpolation of the phono3py BTE results. All quantities dependent on τ are reported scaled by the assumed $\tau = 10^{-14}$ s; absolute magnitudes should be interpreted with the constant- τ approximation in mind.

Fig. S1 — 2D carrier density and optical generation maps

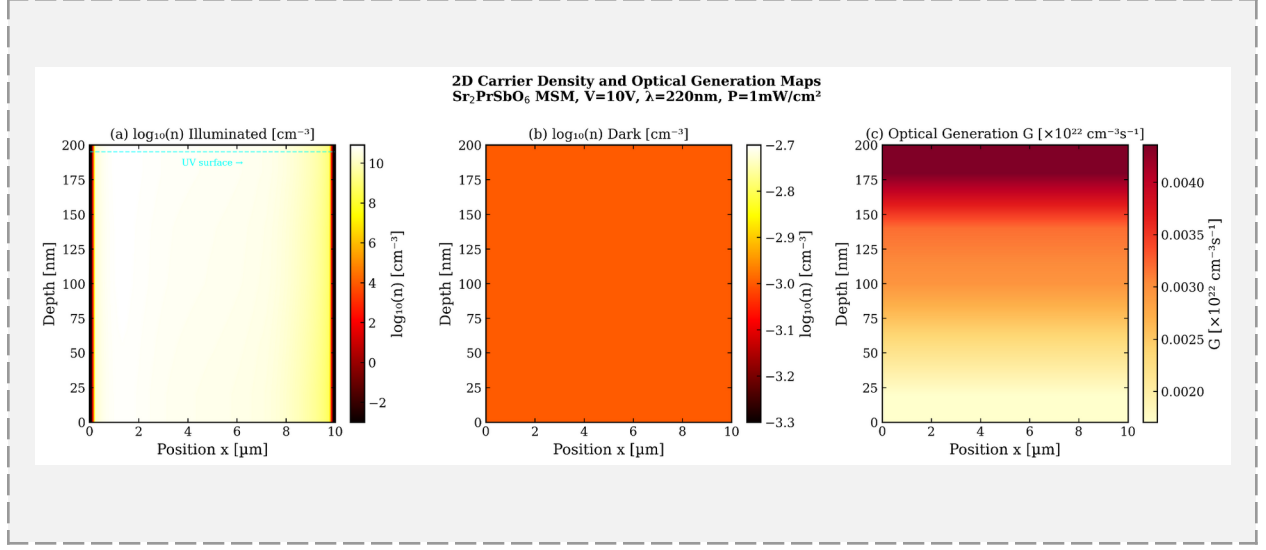


Fig. S1. Two-dimensional spatial maps from the DEVSIMS10 drift-diffusion simulation of the $\text{Sr}_2\text{PrSbO}_6$ metal–semiconductor–metal (MSM) photoconductor ($V = 10 \text{ V}$, $\lambda = 220 \text{ nm}$, $P = 1 \text{ mW/cm}^2$, $T = 300 \text{ K}$, film thickness $d = 200 \text{ nm}$). (a) Log_{10} of the electron density $n \text{ (cm}^{-3}\text{)}$ under UV-C illumination. The laterally uniform carrier density ($\sim 10^{10} \text{ cm}^{-3}$ in the illuminated bulk) drops to its equilibrium value at the contact nodes ($x = 0$ and $x = 10 \text{ } \mu\text{m}$), where the boundary condition fixes the carrier densities. UV-C photons are incident from the top surface (dashed cyan line). (b) Log_{10} of the equilibrium dark electron density (cm^{-3}); the quasi-uniform sub- 10^{-3} cm^{-3} value across the device reflects the vanishingly small intrinsic carrier density arising from the 5.09 eV band gap ($n_i \approx 10^{-24} \text{ cm}^{-3}$). The photocurrent-to-dark-current ratio $I_{ph}/I_d \approx 5.4 \times 10^{13}$ at $V = 10 \text{ V}$ is extracted from the ratio of panels (a) to (b). (c) Volumetric optical generation rate $G \text{ (}\times 10^{22} \text{ cm}^{-3} \text{ s}^{-1}\text{)}$ from the Beer–Lambert model using the HSE06S11,S12 absorption coefficient ($\alpha = 4.69 \times 10^4 \text{ cm}^{-1}$ at 220 nm). The generation profile decreases nearly uniformly with depth because the film thickness ($d = 200 \text{ nm}$) is comparable to the absorption depth ($1/\alpha \approx 213 \text{ nm}$), confirming that the Beer–Lambert approximation of near-uniform generation adopted in the analytical model is self-consistent for this device geometry.

Fig. S2 — 2D electric potential and lateral field maps

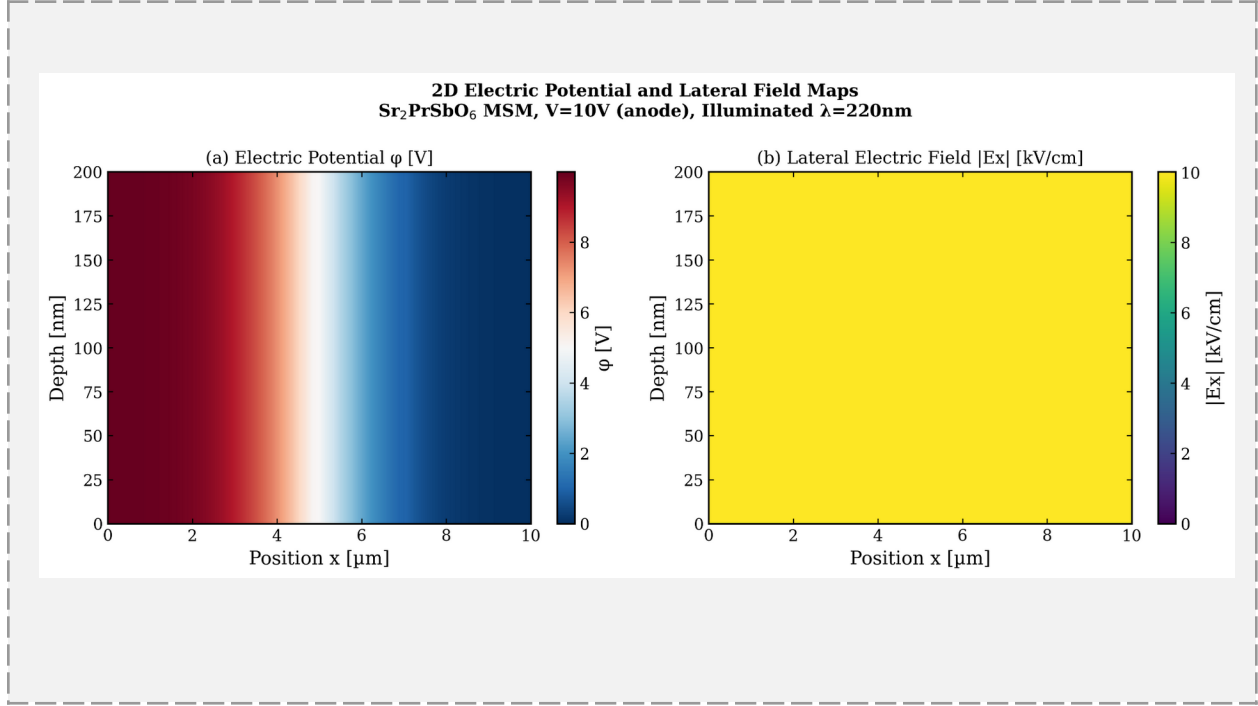


Fig. S2. Two-dimensional electric potential and field distributions in the Sr₂PrSbO₆ MSM photoconductor under UV-C illumination ($V = 10\text{ V}$, $\lambda = 220\text{ nm}$, $T = 300\text{ K}$, $d = 200\text{ nm}$), from DEVSIM TCAD simulation^{S10}. **(a)** Electric potential $\phi(x, y)$ (V), showing the linear drop from the anode ($x = 0$, $\phi = 10\text{ V}$) to the cathode ($x = 10\text{ }\mu\text{m}$, $\phi = 0\text{ V}$), uniform across the film depth. The linear potential profile confirms that the photogenerated carrier density ($\sim 10^{10}\text{ cm}^{-3}$) is too small to cause significant space-charge distortion of the built-in field. **(b)** Lateral electric-field magnitude $|E_x|$ (kV/cm). The field is spatially uniform at $|E_x| = V/L = 10\text{ kV/cm}$ throughout the device interior, as expected for a near-intrinsic photoconductor with ohmic charge transport in the linear I–V regime. The uniformity of both ϕ and $|E_x|$ across the film depth validates the one-dimensional analytical photoconductor model used to compute responsivity and detectivity in the main text.

Fig. S3 — 1D optical generation and lateral carrier profiles

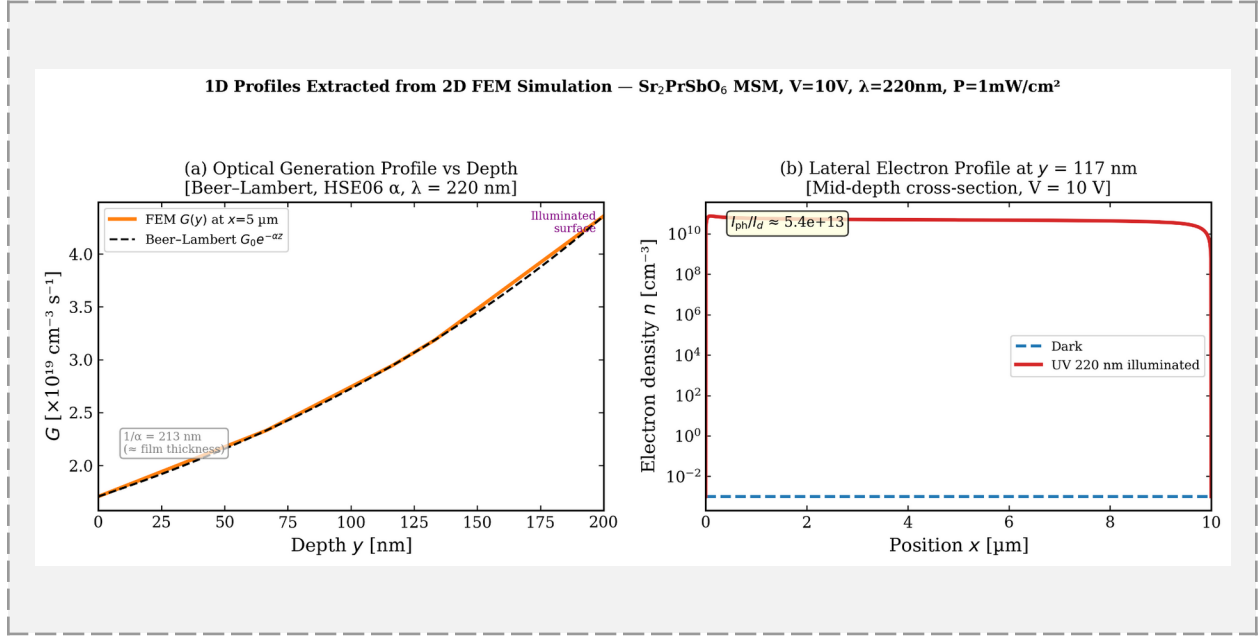
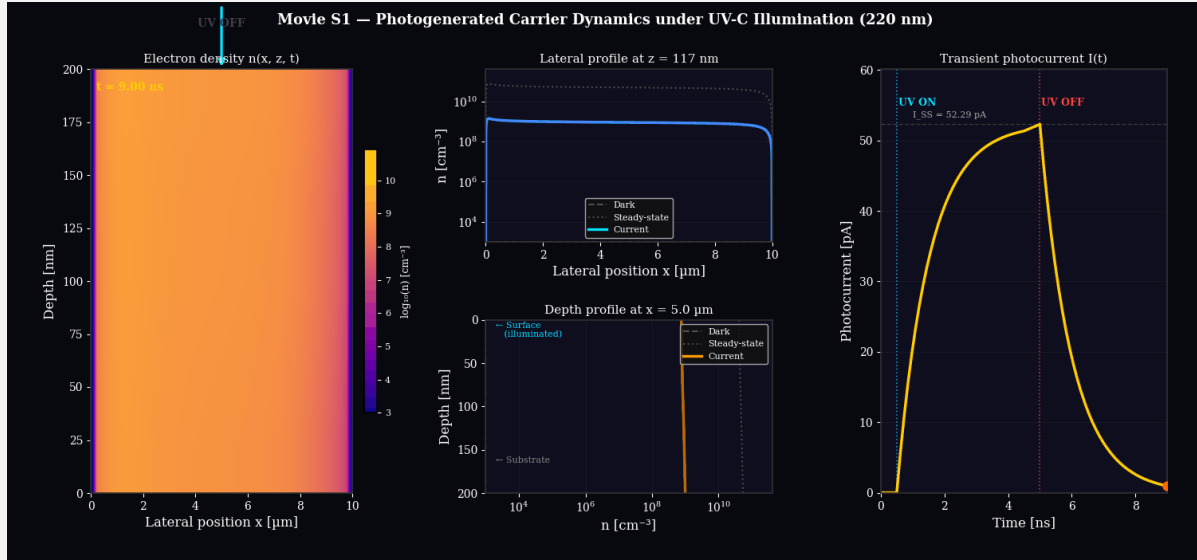


Fig. S3. One-dimensional profiles extracted from the 2D DEVSIM simulation ($V = 10\text{ V}$, $\lambda = 220\text{ nm}$, $P = 1\text{ mW/cm}^2$). (a) Volumetric optical generation rate $G(y)$ as a function of depth y from the illuminated surface, evaluated at the device midpoint ($x = 5\text{ }\mu\text{m}$). The FEM result is in excellent agreement with the Beer-Lambert analytical prediction $G_0 \exp(-\alpha y)$, confirming numerical accuracy. The annotated absorption depth $1/\alpha \approx 213\text{ nm}$ is comparable to the film thickness, so generation is not strictly surface-confined and a mild gradient is present across the 200 nm film. (b) Lateral electron density $n(x)$ at mid-depth ($y = 117\text{ nm}$) for the dark state ($\sim 4 \times 10^{-3} \text{ cm}^{-3}$) and under UV-C illumination ($\sim 10^{10} \text{ cm}^{-3}$). The flat illuminated profile across $0 < x < \sim 9.5\text{ }\mu\text{m}$ confirms bulk photoconductor behaviour with uniform carrier generation; the sharp drop near the cathode contact reflects the contact boundary condition. The labelled ratio $I_{ph}/I_d \approx 5.4 \times 10^{13}$ is consistent with the I-V analysis in the main text.

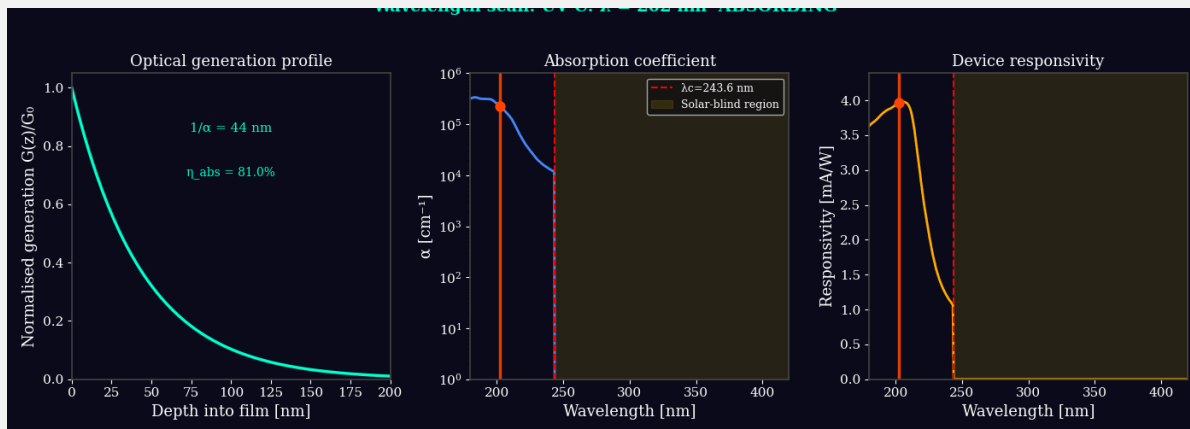
Movie S4: Photogenerated carrier dynamics under UV-C illumination

Time-domain simulation of electron-hole pair generation and drift under 220 nm UV-C illumination (1 mW/cm^2 , $V = 10\text{ V}$). Carriers are generated non-uniformly following the Beer-Lambert absorption profile. The animation shows the exponential rise ($\tau_{\text{rise}} \approx 1\text{ ns}$) as carriers build up to steady state and the subsequent exponential decay ($\tau_{\text{fall}} \approx 1\text{ ns}$) when illumination is removed, demonstrating the intrinsic SRH recombination-limited response time^{S13}.



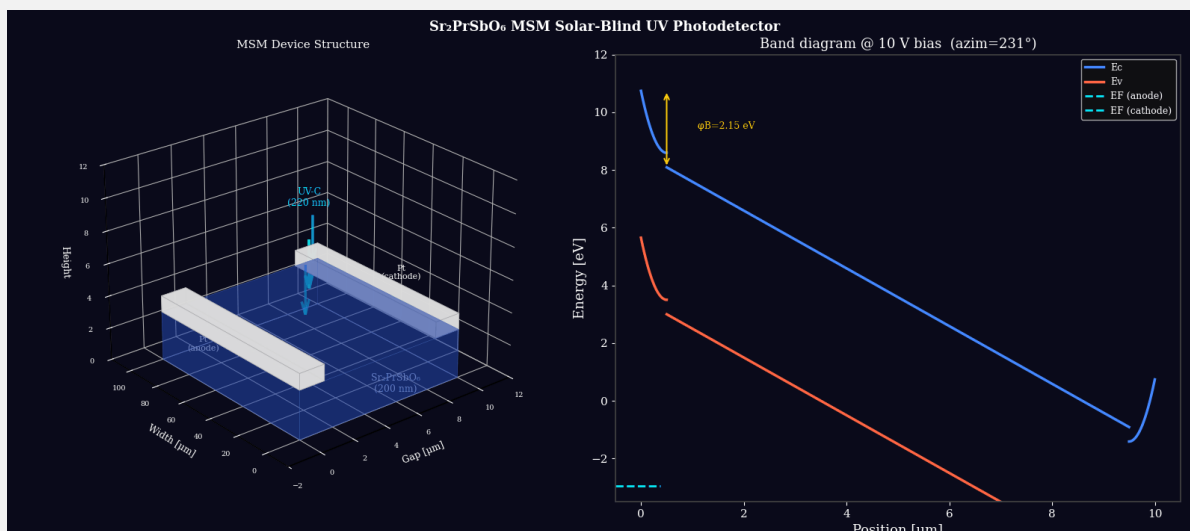
Movie S5: Absorption and generation profile as a function of wavelength (180–400 nm)

This animation demonstrates the solar-blind mechanism by sweeping the illumination wavelength from 180 nm (deep UV-C) to 400 nm (visible). The generation profile $G(z, \lambda) = G_0(1 - R)\exp(-\alpha z)$ evolves dramatically: strong absorption with $1/\alpha \approx 20$ nm penetration depth at 200 nm transitions to zero generation at the solar-blind cutoff $\lambda_c = 243.6$ nm (corresponding to the HSE06 band gap $E_g = 5.09$ eV). For all wavelengths $\lambda > 243.6$ nm — including the entire visible solar spectrum — the absorption coefficient is identically zero, ensuring complete rejection of solar background radiation.



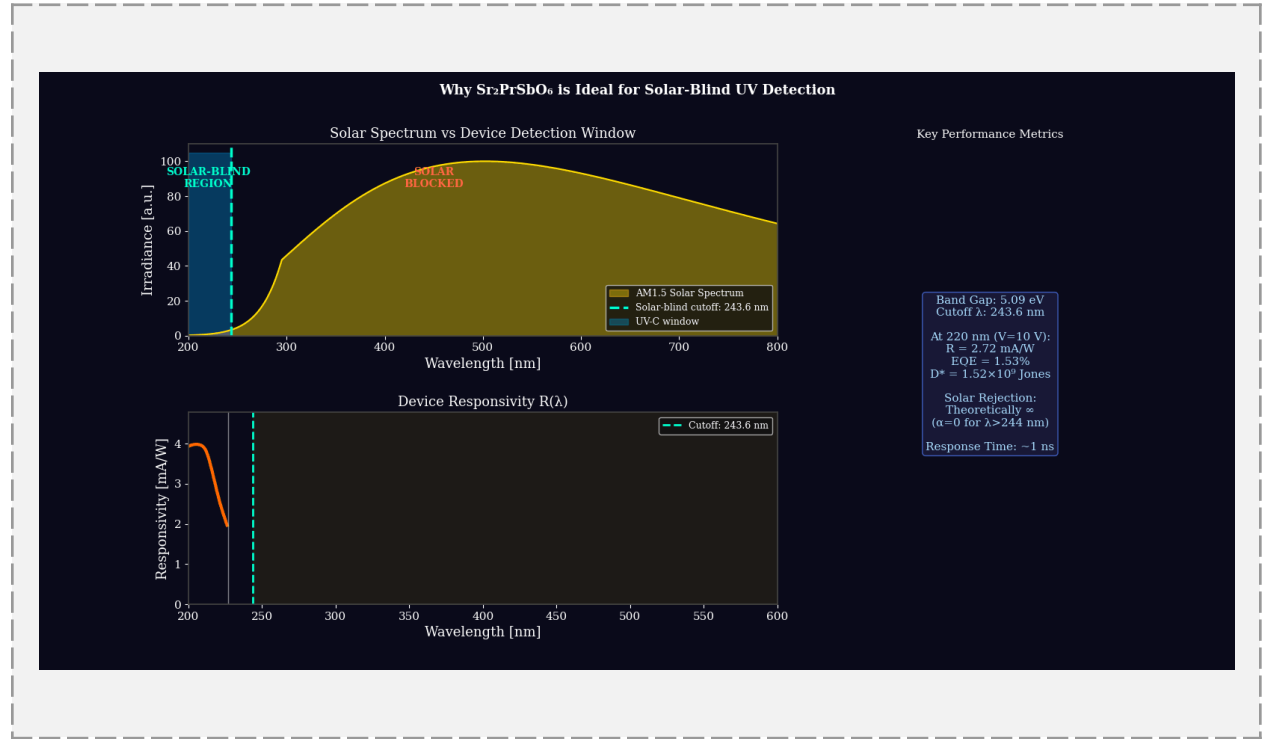
Movie S6: Three-dimensional MSM device structure and band diagram

Rotating three-dimensional visualization of the $\text{Pt}/\text{Sr}_2\text{PrSbO}_6/\text{Pt}$ MSM photoconductor geometry alongside the energy band diagram. The semiconductor film (200 nm thick, 10 μm gap, 100 μm wide) is sandwiched between two Pt finger electrodes. The band diagram shows the characteristic double back-to-back Schottky barrier profile under 10 V bias, with 2.15 eV barriers at each metal–semiconductor interface. UV-C photons enter from the top surface, creating electron–hole pairs that are swept laterally toward opposite contacts by the applied field.



Movie S7: Solar-blind operation demonstrated against the AM1.5 solar spectrum

This animation illustrates the fundamental advantage of $\text{Sr}_2\text{PrSbO}_6$ for solar-blind UV sensing: the device responsivity $R(\lambda)$ and the AM1.5 solar spectrum have zero spectral overlap. The wide band gap of 5.09 eV (HSE06) places the detection window exclusively in the UV-C region ($\lambda < 243.6$ nm), where solar irradiance at Earth's surface is effectively zero due to stratospheric ozone absorption. This eliminates the need for optical filters — the device is intrinsically solar-blind — and the theoretical UV/visible rejection ratio exceeds 10^{15} , limited only by the sharpness of the HSE06-calculated absorption edge.



Movie S8: Three-dimensional responsivity landscape $R(\lambda, V)$

The three-dimensional responsivity surface $R(\lambda, V)$ reveals the optimization landscape of the device. Responsivity peaks near the band-gap edge ($\lambda \approx 220\text{--}240$ nm) and increases monotonically with bias voltage as field-enhanced carrier sweep-out improves collection efficiency. The sharp truncation at $\lambda_c = 243.6$ nm confirms the solar-blind cutoff, beyond which $R = 0$ for all bias values. The recommended operating point of $\lambda = 220$ nm and $V = 10$ V (marked) provides a favourable trade-off between sensitivity and device stability.

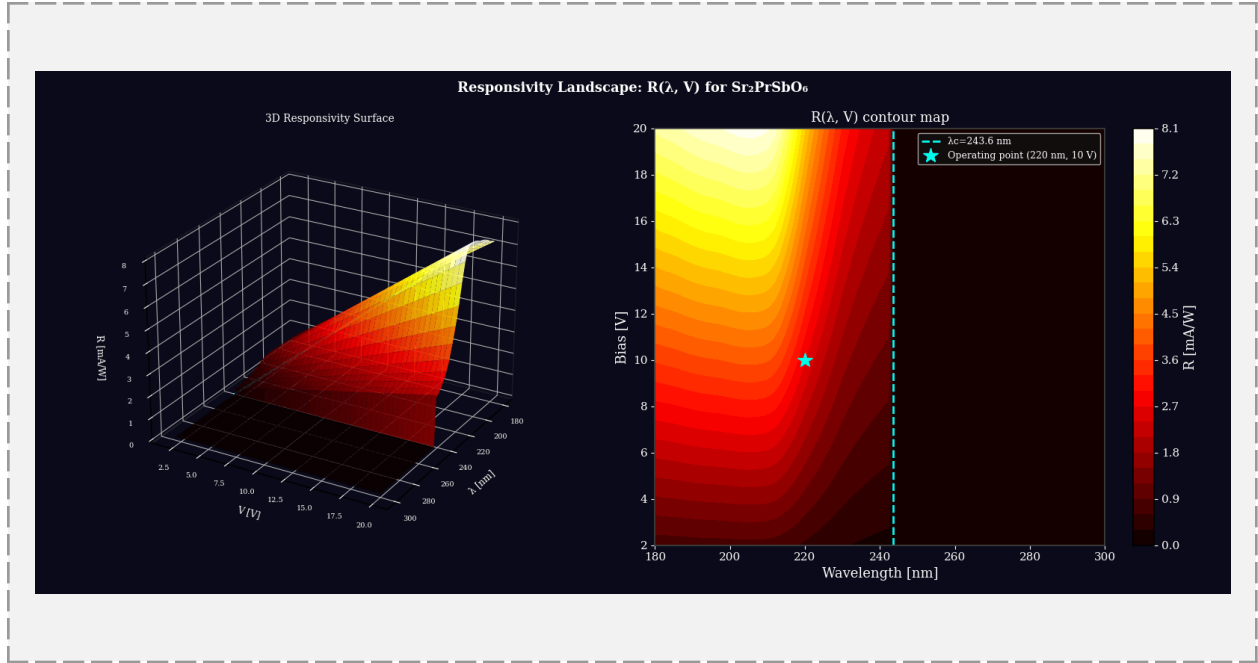


Table S9 — Complete simulation parameter set

Table S9. Complete simulation parameter set for the $\text{Sr}_2\text{PrSbO}_6$ MSM photoconductor FEM simulation. Parameters are categorized by source.

Parameter	Symbol	Value	Unit	Source	Note
Band gap	E_g	5.09	eV	This work	HSE06 gives indirect gap
Absorption onset	λ_c	243.6	nm	Derived	solar-blind UV-C
Static dielectric const.	ϵ_r	3.5	—	This work	$\epsilon_1(0)$, averaged over tensor components
Electron mobility	μ_e	2.0	cm^2/Vs	This work	Calculated using Bardeen–Shockley deformation-potential theory
Hole mobility	μ_h	1.0	cm^2/Vs	This work	Calculated using Bardeen–Shockley deformation-potential theory
SRH carrier lifetime	τ	1	ns	Literature ^{S13}	wide-gap oxide thin-film photodetectors
Electron affinity	χ	3.5	eV	This work	Calculated using VASP
Electron effective mass	m_e^*	0.58	m_0	This work	parabolic fit of the HSE06 conduction band at Γ (Table S16)
Hole effective mass	m_h^*	2.7	m_0	This work	parabolic fit of the HSE06 valence band at D (Table S16)
Schottky barrier (e^-)	Φ_B	2.15	eV	Derived	$\Phi_{\text{Pt}} - \chi$ (Schottky–Mott limit); 2.25 eV in the Bardeen (pinned) limit, $E_C - E_{\text{BP}}$

Parameter	Symbol	Value	Unit	Source	Note
Pt work function	Φ_{Pt}	5.65	eV	Literature ^{S15}	Michaelson 1977; standard reference value
Dark current density	J_d	10^{-10}	A/cm ²	Literature ^{S16}	β -Ga ₂ O ₃ MSM reference
Film thickness (baseline)	d	200	nm	Device design	sensitivity study: 200–5000 nm
Electrode gap	L	10	μm	Device design	MSM spacing
Device width	W	100	μm	Device design	—
Temperature	T	300	K	Standard	room temperature
Branch-point energy	E_{BP}	2.84	eV	This work	above the VBM; HSE06, Supplementary Note 3
Barrier height, Bardeen limit	$E_{\text{C}} - E_{\text{BP}}$	2.25	eV	Derived	pinned-interface limit; Section 3.5.1

Table S10 — Murnaghan equation-of-state parameters

Table S10. Murnaghan equation-of-state parameters for Sr₂PrSbO₆ computed for the cubic Fm-3m aristotype cell (40 atoms) (GGA-PBES8).

Parameter	Value	Description
V_0 (Å ³ /atom)	15.894	GGA-PBE equilibrium volume per atom
V_0 (Å ³ /cell)	317.9	Conventional cell volume
E_0 (eV/atom)	−6.704	Equilibrium total energy
B_0 (GPa)	125.5	Zero-pressure bulk modulus

Supplementary Note 1 — Thermoelectric transport properties of Sr₂PrSbO₆

Figure S11 presents the temperature dependence of the lattice thermal conductivity resolved along the three crystallographic axes. At 300 K the transport is markedly anisotropic: $\kappa_{xx} = 49.42$, $\kappa_{yy} = 42.95$, and $\kappa_{zz} = 32.63 \text{ W m}^{-1} \text{ K}^{-1}$, giving an average $\kappa_{\text{avg}} = 41.67 \text{ W m}^{-1} \text{ K}^{-1}$ and an anisotropy ratio $\kappa_{xx}/\kappa_{zz} = 1.52$. The thermal conductivity shows a broad maximum near 400 K, arising from the interplay between thermally activated optical-phonon occupancy and enhanced three-phonon Umklapp scattering, and decreases approximately as T^{-1} to $22.26 \text{ W m}^{-1} \text{ K}^{-1}$ at 1100 K (a 47% reduction over 400–1100 K). Mode decomposition [Fig. S11(b)] reveals that optical branches carry 99.6% of the heat at 300 K ($\kappa_{\text{op}} = 41.50$ vs. $\kappa_{\text{ac}} = 0.168 \text{ W m}^{-1} \text{ K}^{-1}$), an unusual inversion of the acoustic-dominated transport typical of crystalline

solids but consistent with reports of optical-mode dominance and wave-like phonon tunnelling in complex double perovskites^{S17}.

The spectral decomposition [Fig. S12(a)] confirms that heat transport is dominated by modes between 13–20 THz, with a prominent peak at ~19.5 THz corresponding to the highest-lying flat optical branches of SbO_6 stretching character; the spectral shape is nearly temperature-independent, though absolute magnitudes decrease at higher temperatures. The cumulative mean-free-path (MFP) analysis [Fig. S12(b)] shows that at 300 K, 50% of the total thermal conductivity is carried by phonons with MFP < 3.9 μm , and 90% by phonons with MFP < 12.0 μm .

The electronic transport coefficients as functions of chemical potential $\mu - E_F$ are presented in Fig. S13 for temperatures of 300–1100 K. The Seebeck coefficient is large and positive (p-type behaviour) for $-1.5 < \mu - E_F < -0.1$ eV, reaching 300–400 $\mu\text{V K}^{-1}$ near the valence band, indicating hole-dominated transport at these doping levels. The power factor $S^2\sigma$ peaks at $\mu - E_F \approx -0.141$ eV, where the Seebeck coefficient remains large while the conductivity is enhanced by proximity to the valence band; the peak power factor increases from 1.31 $\text{mW K}^{-2} \text{m}^{-1}$ at 300 K to 3.01 $\text{mW K}^{-2} \text{m}^{-1}$ at 1100 K, values comparable to moderate oxide thermoelectrics^{S18,S19}. The optimal chemical potential is essentially temperature-independent.

Figure S14 summarizes the temperature dependence of all thermoelectric properties evaluated at $\mu_{\text{opt}} - E_F = -0.141$ eV. The Seebeck coefficient remains nearly constant at 185–200 $\mu\text{V K}^{-1}$ over 300–1100 K, the electrical conductivity increases monotonically, and the electronic thermal conductivity κ_e is at most 1.49 $\text{W m}^{-1} \text{K}^{-1}$ at 1100 K, confirming that $\kappa_L \gg \kappa_e$ and that lattice conduction dominates throughout. The maximum figure of merit grows from $ZT = 0.0094$ at 300 K to 0.14 at 1100 K, driven primarily by the reduction of κ_L with temperature rather than by changes in the electronic power factor. While $\text{Sr}_2\text{PrSbO}_6$ is not competitive with state-of-the-art thermoelectric materials as a standalone energy converter, these characteristics are directly relevant to the photodetector application: the high thermal conductivity facilitates heat dissipation under UV irradiation, and the appreciable Seebeck response may be exploited for self-powered or thermally modulated sensing schemes.

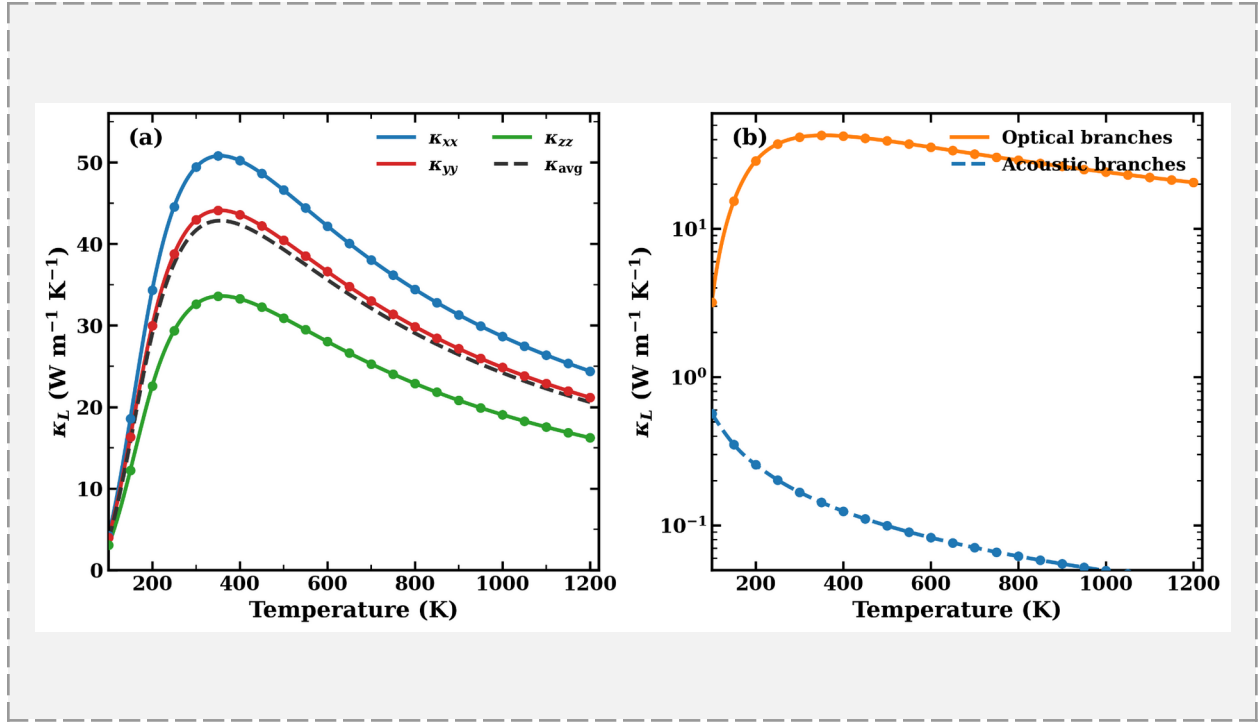


Fig. S11. Lattice thermal conductivity of $\text{Sr}_2\text{PrSbO}_6$ vs. temperature. **(a)** Anisotropic components κ_{xx} , κ_{yy} , κ_{zz} , and the average κ_{avg} computed by the phono3py BTE^{S1–S3}. Points are raw BTE results; curves are cubic-spline interpolations. **(b)** Decomposition into acoustic and optical branch contributions (log scale).

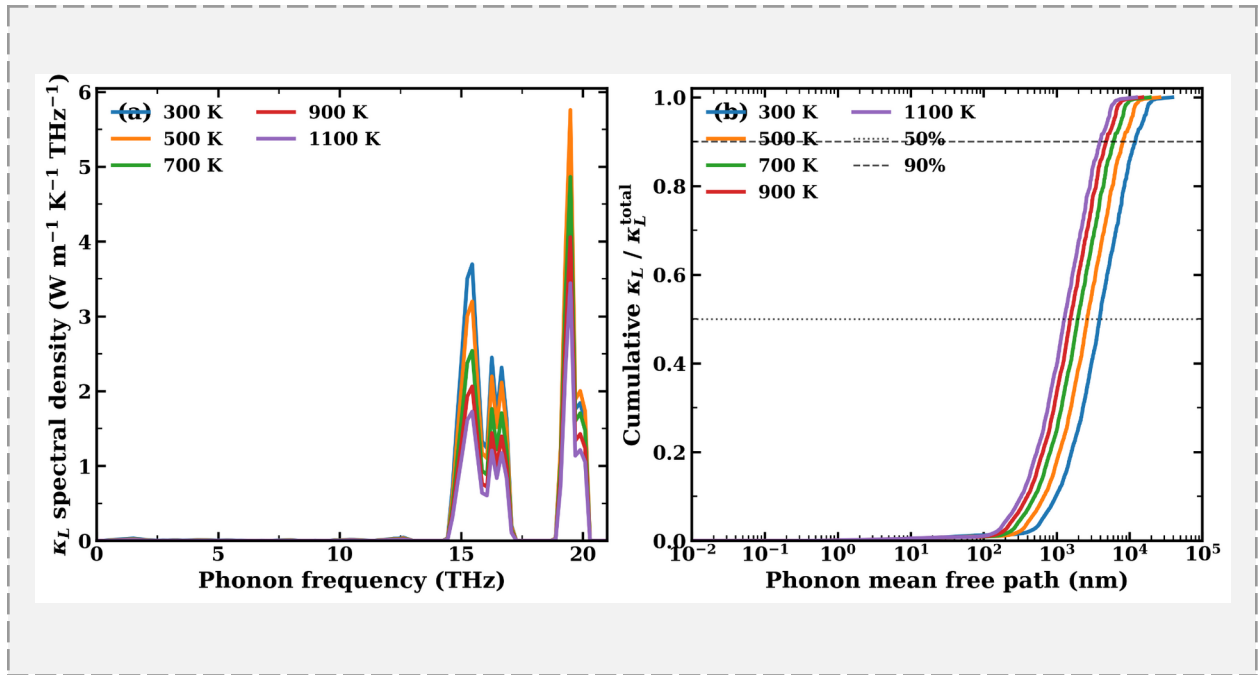


Fig. S12. **(a)** Spectral lattice thermal conductivity density $\kappa_L(\nu)$ as a function of phonon frequency for temperatures of 300–1100 K. **(b)** Cumulative κ_L , normalized to the total, as a function of phonon mean

free path (MFP). Horizontal reference lines mark the 50% and 90% levels. At 300 K, 50% of the heat is carried by phonons with MFP < 3.9 μm .

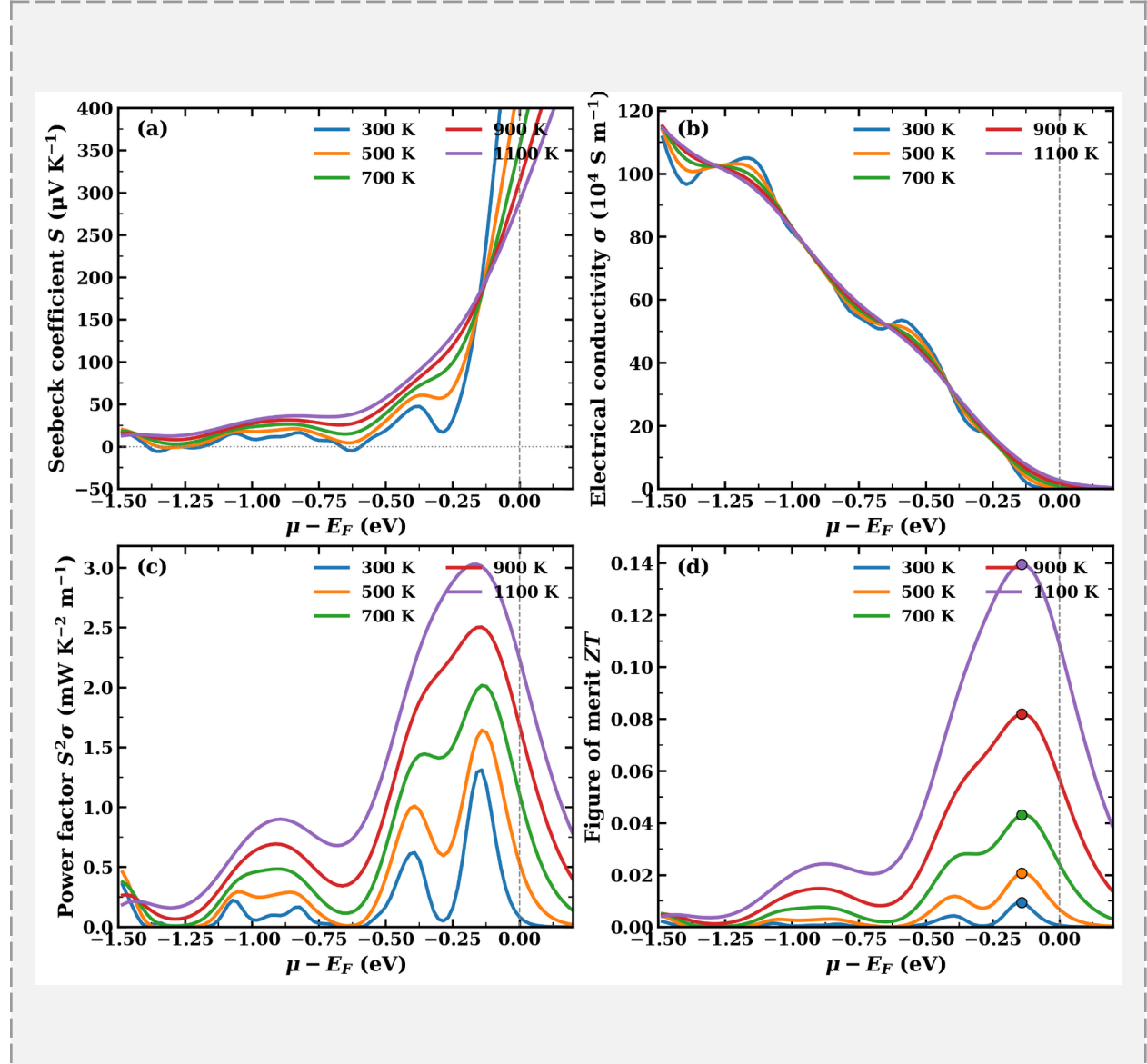


Fig. S13. Electronic transport properties of $\text{Sr}_2\text{PrSbO}_6$ as functions of chemical potential $\mu - E_F$ at temperatures of 300–1100 K (constant relaxation time $\tau = 10^{-14}$ s, BoltzTraP^{S9}). (a) Seebeck coefficient S . (b) Electrical conductivity σ . (c) Power factor $S^2\sigma$. (d) Figure of merit ZT ; markers indicate the maximum ZT at each temperature.

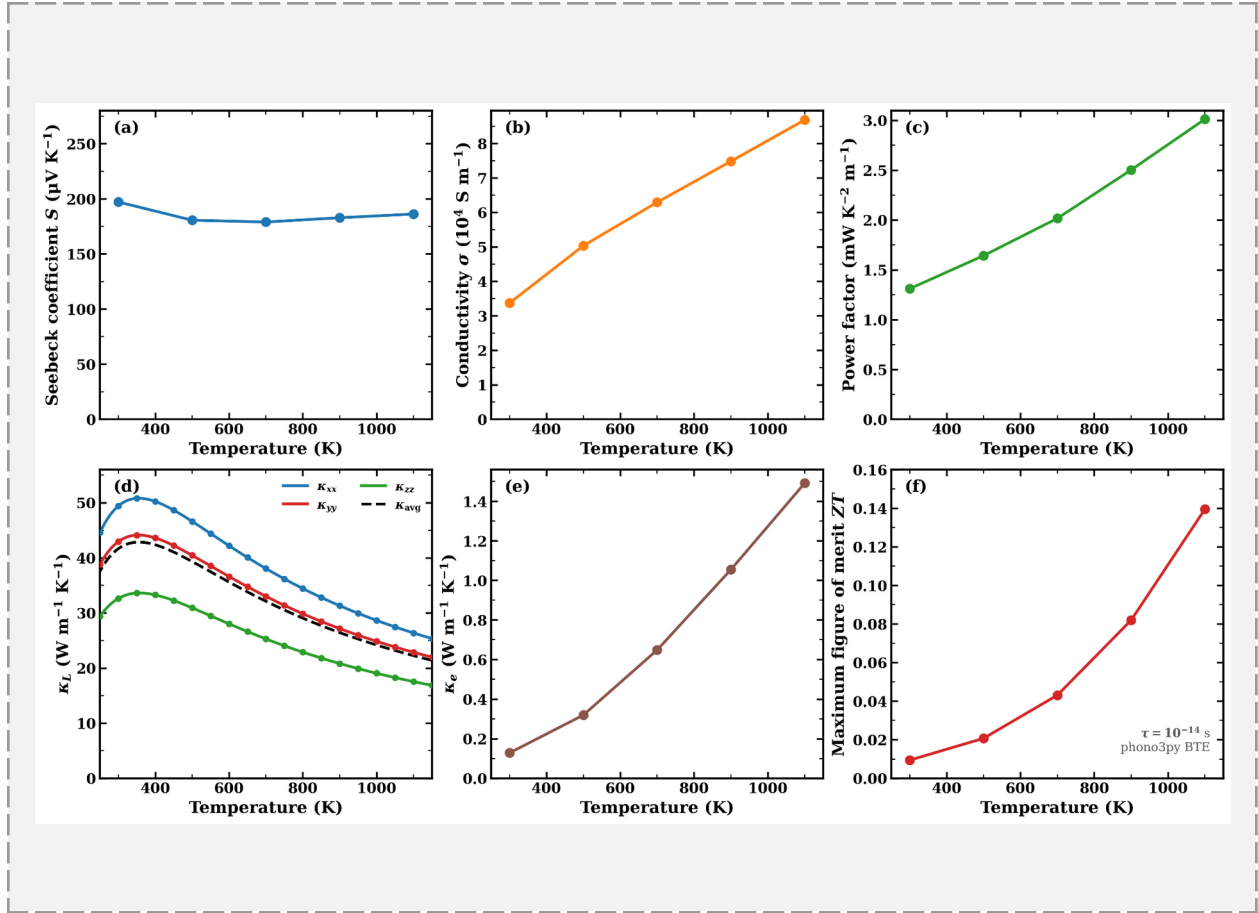


Fig. S14. Temperature dependence of thermoelectric properties at the optimal chemical potential $\mu_{\text{opt}} - E_{\text{F}} = -0.141$ eV. **(a)** Seebeck coefficient. **(b)** Electrical conductivity. **(c)** Power factor. **(d)** Anisotropic lattice thermal conductivity κ_{xx} , κ_{yy} , κ_{zz} , κ_{avg} . **(e)** Electronic thermal conductivity κ_e . **(f)** Maximum figure of merit ZT ($\tau = 10^{-14}$ s; phono3py BTE κ_L).

Supplementary Note 2 — Mechanical properties of $\text{Sr}_2\text{PrSbO}_6$

The mechanical response of $\text{Sr}_2\text{PrSbO}_6$ was characterised from the full 6×6 elastic stiffness tensor of the monoclinic $P2_1/c$ structure, which contains 13 independent components C_{ij} , computed by the finite-difference method within the GGA-PBE functional^{S8} and post-processed with VASPKIT^{S20}. The diagonal elements $C_{11} = 191.8$ GPa, $C_{22} = 202.8$ GPa, and $C_{33} = 221.7$ GPa indicate strong resistance to uniaxial compression in all three crystallographic directions. The off-diagonal stiffness constants $C_{12} = 102.4$ GPa and $C_{13} = 70.1$ GPa indicate significant cross-directional coupling, and the shear moduli $C_{44} = 60.5$ GPa, $C_{55} = 51.1$ GPa, and $C_{66} = 75.9$ GPa confirm resistance to shear deformation.

The polycrystalline elastic moduli obtained from the Voigt–Reuss–Hill (VRH) averaging scheme are summarised in Table S15. The Hill averages are: bulk modulus $B_{\text{H}} = 121.6$ GPa, shear modulus $G_{\text{H}} = 60.9$ GPa, Young's modulus $E_{\text{H}} = 156.5$ GPa, and Poisson's ratio $\nu_{\text{H}} = 0.285$. The Pugh ratio $B/G = 2.00$

exceeds the empirical ductile/brittle threshold of 1.75^{S21}, classifying Sr₂PrSbO₆ as a ductile oxide. The Vickers hardness estimated using the Chen^{S22} and Tian^{S23} empirical relations yields H_V = 6.8 and 7.7 GPa, respectively. The Cauchy pressure C₁₂ – C₄₄ = 41.9 GPa is positive, indicating mixed ionic–covalent bonding character^{S24}. The Debye temperature derived from the elastic wave velocities (v_L = 5915 m/s, v_T = 3241 m/s) is θ_D = 433.4 K. Together with the satisfaction of the Born–Huang stability criteria discussed in the main text, these moduli confirm that Sr₂PrSbO₆ is mechanically robust for thin-film device applications.

Table S15 — Polycrystalline elastic moduli

Table S15. Calculated polycrystalline elastic moduli and related quantities for Sr₂PrSbO₆ from the Voigt–Reuss–Hill (VRH) averaging scheme (GGA-PBE^{S8}).

Property	Voigt	Reuss	Hill
Bulk modulus B (GPa)	121.61	121.51	121.56
Shear modulus G (GPa)	62.65	59.09	60.87
Young's modulus E (GPa)	160.41	152.54	156.49
Poisson's ratio ν	0.280	0.291	0.285
P-wave modulus M (GPa)	205.14	200.30	202.72
Pugh's ratio B/G	1.94	2.06	2.00
Vickers hardness H _V (Chen ^{S22}) (GPa)	7.30	6.31	6.80
Vickers hardness H _V (Tian ^{S23}) (GPa)	8.10	7.28	7.69
Cauchy pressure P _c (GPa)	—	—	41.9
Universal anisotropy index A ^U	—	—	0.30
Debye temperature θ _D (K)	—	—	433.4
Longitudinal velocity v _L (m/s)	—	—	5915
Transverse velocity v _T (m/s)	—	—	3241

Supplementary Note 3 — Branch-point energy and band-edge effective masses of Sr₂PrSbO₆

The branch-point energy (charge-neutrality level) E_{BP} was evaluated from the HSE06 eigenvalues as the Brillouin-zone average of the mid-point between the N_{CB} lowest conduction bands and the N_{VB} highest valence bands, E_{BP} = (1/2N_k) Σ_k [(1/N_{CB}) Σ_i ε_{c,i}(k) + (1/N_{VB}) Σ_j ε_{v,j}(k)], following Tersoff^{S25} and Schleife et al.^{S26} The weighted average was taken over the 24 symmetry-irreducible points of the Γ-centred 5×4×3 mesh of the HSE06 band-structure calculation (the 124 zero-weight k-path points were used only as a cross-check); N_{CB} = 2 and N_{VB} = 4 were adopted for the two-formula-unit cell, and the sensitivity to this choice is shown in Table S16. Band-edge effective masses were obtained by parabolic

fits $E(\mathbf{k}) = E_0 + \hbar^2 \mathbf{k}^2 / 2m^*$ to the HSE06 bands over the first four k-path points ($|\mathbf{k}| \leq 0.14 \text{ \AA}^{-1}$) from each extremum.

E_{BP} lies 2.84 eV above the valence-band maximum and 2.25 eV below the conduction-band minimum (0.56 of the gap), robust to the band averaging (2.84–3.11 eV) and essentially unchanged when evaluated along the k-path (2.82 eV). A branch point deep inside the gap classifies $\text{Sr}_2\text{PrSbO}_6$ as an intrinsically insulating oxide with at most weak n-type and no realistic p-type dopability,^{S26,S27} in line with the nearly dispersionless O-2p valence band (bandwidth 0.17 eV, $m_h^* \approx 2.6\text{--}2.8 m_0$) characteristic of hole self-trapping in wide-gap oxides^{S28} and the light, dispersive Sb-5s/Pr-5d conduction band (bandwidth 1.71 eV, $m_e^* = 0.53\text{--}0.64 m_0$). In the Bardeen (fully pinned) limit the Pt barrier height is $\Phi_{B,n} = E_C - E_{BP} = 2.25 \text{ eV}$, compared with 2.15 eV in the Schottky–Mott limit (Section 3.5.1 of the main text).

Table S16 — Branch-point energy and band-edge effective masses

Table S16. Branch-point energy E_{BP} of $\text{Sr}_2\text{PrSbO}_6$ from the HSE06 band structure for different numbers of averaged conduction (N_{CB}) and valence (N_{VB}) bands, and band-edge effective masses from parabolic fits along the indicated directions of the monoclinic Brillouin zone. $E_g = 5.090 \text{ eV}$ (VBM at D, CBM at Γ).

Quantity	Value	Note
$E_{BP} - E_{VBM}, (N_{CB}, N_{VB}) = (1, 2)$	2.853 eV	$(E_{BP} - E_{VBM})/E_g = 0.560$
$E_{BP} - E_{VBM}, (N_{CB}, N_{VB}) = (2, 4)$	2.838 eV	reference choice ($Z = 2 \text{ f.u.}$); $E_{CBM} - E_{BP} = 2.252 \text{ eV}$; 0.558 of the gap
$E_{BP} - E_{VBM}, (N_{CB}, N_{VB}) = (3, 6)$	2.974 eV	0.584 of the gap
$E_{BP} - E_{VBM}, (N_{CB}, N_{VB}) = (4, 8)$	3.033 eV	0.596 of the gap
$E_{BP} - E_{VBM}, (N_{CB}, N_{VB}) = (6, 12)$	3.091 eV	0.607 of the gap
$E_{BP} - E_{VBM}$, k-path average, (2, 4)	2.817 eV	unweighted average over the 124 band-structure k-points
Mid-gap reference	2.545 eV	$E_g/2$
m_e^* (CBM at Γ): $\Gamma \rightarrow Z, \Gamma \rightarrow B, \Gamma \rightarrow A, \Gamma \rightarrow Y_2$	0.560, 0.528, 0.636, 0.593 m_0	harmonic mean 0.58 m_0 ; conduction-band width 1.71 eV
m_h^* (VBM at D): $D \rightarrow Z, D \rightarrow B$	2.84, 2.64 m_0	harmonic mean 2.7 m_0 ; valence-band width 0.17 eV
CBM character (PROCAR, Γ)	Sb-5s 34 %, Pr-5d 29 %, O-2p/2s 32 %, Sr-4d 4 %	Pr 4f ² treated as frozen core (Pr_3 potential)
VBM character (PROCAR, D)	O-2p 97 %	—

Supplementary Note 4 — Formation enthalpy and temperature-dependent stability of $\text{Sr}_2\text{PrSbO}_6$

The formation enthalpy $\Delta H_f = E(\text{Sr}_2\text{PrSbO}_6) - [2E(\text{Sr}) + E(\text{Pr}) + E(\text{Sb}) + 3E(\text{O}_2)]$ was evaluated at $T = 0$ K in the static-lattice approximation from GGA-PBE^{S8} total energies computed with a 600 eV cutoff and identical PAW potentials (Sr_sv, Pr_3, Sb, O) for the compound (PBE-relaxed monoclinic cell, 7×5×4 mesh) and for the elemental ground states: fcc Sr ($a = 6.023$ Å), double-hexagonal close-packed α -Pr ($a = 3.744$, $c = 12.02$ Å), rhombohedral A7 Sb (31.8 Å³/atom) and the spin-triplet O₂ molecule ($d = 1.233$ Å) in a 12 Å box, each fully relaxed. Because PBE over-binds O₂, ΔH_f is also given with the fitted oxygen correction of -0.687 eV per O atom (Materials Project MP2020 scheme).^{S29,S30} The zero-point energy of the compound (52.7 meV/atom) was taken from the phonon calculation and those of the elements from Debye models (O₂: $\frac{1}{2}\hbar\omega_e$ with $\omega_e = 1580.19$ cm⁻¹^{S35}). The Gibbs free energy of formation at 1 bar, $\Delta G_f(T) = \Delta H_f(0 \text{ K}) + \Delta F_{\text{vib}}(T)$, combines the harmonic vibrational free energy of Sr₂PrSbO₆ (Phonopy,^{S1} 2×2×1 supercell, 16×16×16 q-mesh) with Debye-model free energies of the metals ($\theta_D = 147, 152$ and 211 K for Sr, Pr and Sb^{S33,S34}) and the NIST-JANAF enthalpy and entropy of O₂(g).^{S31} The Debye treatment of the metals was checked against the NIST-JANAF entropy of α -Sr (51.0 vs. 55.7 J K⁻¹ mol⁻¹ at 298 K), which bounds its contribution to ΔG_f to ≈ 0.01 eV/atom.

The results are collected in Fig. S17 and Table S18: $\Delta H_f = -2.61$ eV/atom (-2518 kJ mol⁻¹) without and -3.02 eV/atom with the O₂ correction, values that bracket the sum of the experimental 298 K formation enthalpies of the constituent binary oxides, $2 \text{ SrO} + \frac{1}{2} \text{ Pr}_2\text{O}_3 + \frac{1}{2} \text{ Sb}_2\text{O}_5 = -2.67$ eV/atom,^{S31,S32} which is a lower bound on $|\Delta H_f|$ of the double perovskite. ΔG_f becomes less negative with temperature because of the translational entropy of the three O₂ molecules released per formula unit ($\Delta S_f \approx -0.52$ to -0.56 kJ K⁻¹ mol⁻¹), but remains -2.43 eV/atom at 300 K and -2.04 eV/atom at 1000 K. The thermodynamic stability of Sr₂PrSbO₆ is also established experimentally: the compound has been synthesised by solid-state reaction and refined in P2₁/n,^{S36} the HSE06 cell of Table 1 (main text) reproducing the experimental lattice parameters to within 1 %, and the isostructural Sr₂LnSbO₆ antimonates are prepared in air at 1250–1350 °C and show only displacive octahedral-tilt transitions (P2₁/n $\rightarrow$ R-3 $\rightarrow$ Fm-3m at ≈ 885 – 945 K and ≈ 1170 – 1180 K for Ln = Sm, La) without decomposition.^{S37,S38}

Fig. S17 — Thermodynamic functions and Gibbs free energy of formation

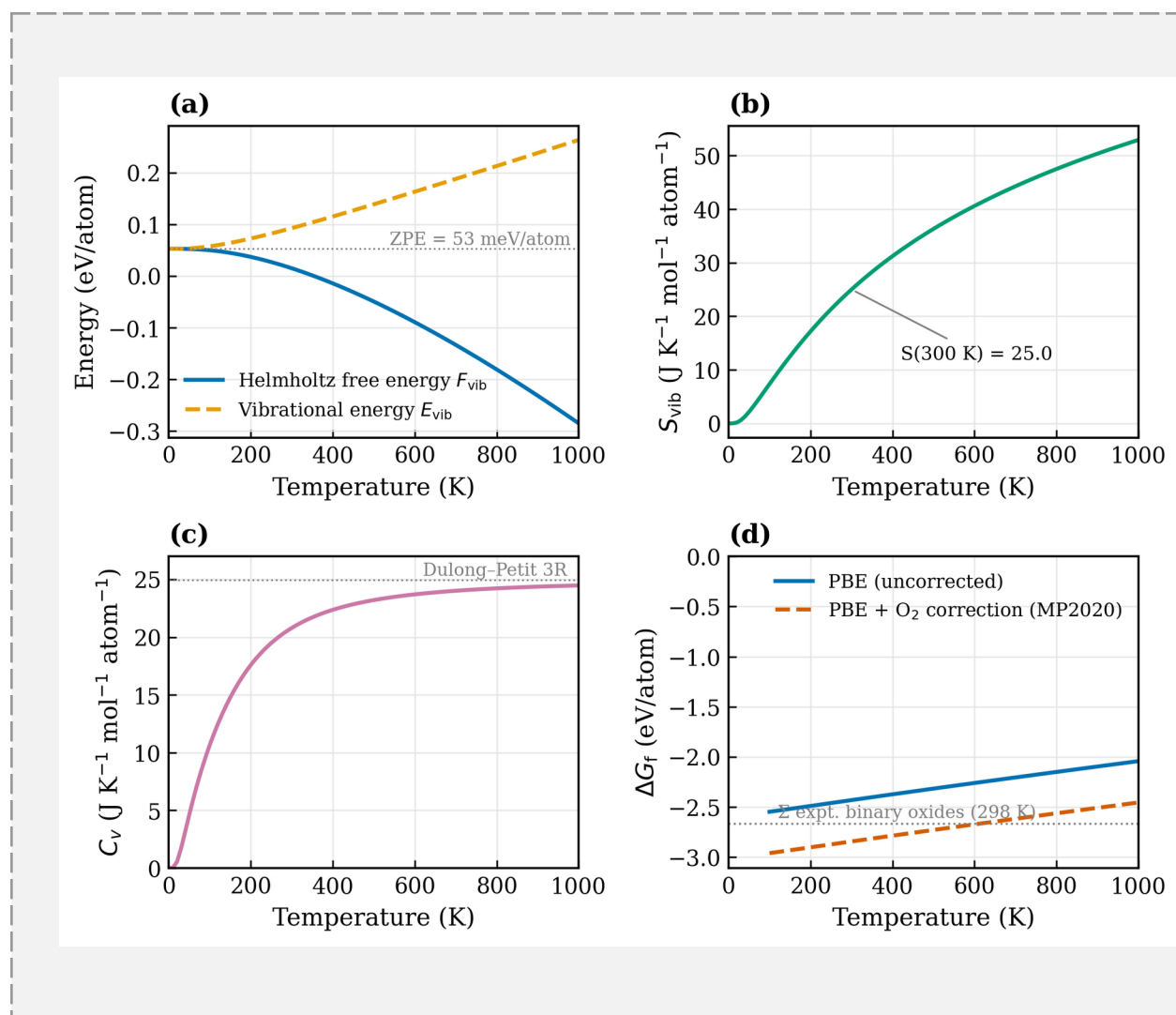


Fig. S17. Harmonic thermodynamic functions of $\text{Sr}_2\text{PrSbO}_6$ (Phonopy, $2 \times 2 \times 1$ supercell) and Gibbs free energy of formation. (a) Vibrational Helmholtz free energy F_{vib} and internal energy E_{vib} per atom (zero-point energy 53 meV/atom). (b) Vibrational entropy. (c) Constant-volume heat capacity, approaching the Dulong–Petit limit $3R$. (d) $\Delta G_f(T)$ relative to the elemental ground states at 1 bar from uncorrected PBE (solid) and with the MP2020 O_2 correction (dashed); metals treated in the Debye model, $\text{O}_2(\text{g})$ from NIST-JANAF. The dotted line marks the sum of the experimental 298 K formation enthalpies of SrO , Pr_2O_3 and Sb_2O_5 .

Table S18 — Formation enthalpy and Gibbs free energy of formation

Table S18. Formation enthalpy and Gibbs free energy of formation of $\text{Sr}_2\text{PrSbO}_6$ (GGA-PBE, 600 eV; elemental references in their ground-state structures; standard state $p(\text{O}_2) = 1$ bar).

Quantity	Value
ΔH_f (0 K, static lattice, PBE)	$-26.10 \text{ eV/f.u.} = -2.610 \text{ eV/atom} = -2518 \text{ kJ mol}^{-1}$

ΔH_f including zero-point energy	−2.593 eV/atom
ΔH_f with O ₂ correction (MP2020, −0.687 eV per O atom)	−3.022 eV/atom
Σ experimental ΔH_f° (298 K), 2 SrO + ½ Pr ₂ O ₃ + ½ Sb ₂ O ₅	−2.669 eV/atom (−2575 kJ mol ^{−1})
ΔG_f at 100 / 300 / 500 / 700 / 1000 K (uncorrected)	−2.55 / −2.43 / −2.32 / −2.20 / −2.04 eV/atom
ΔG_f at 100 / 300 / 500 / 700 / 1000 K (O ₂ -corrected)	−2.96 / −2.84 / −2.73 / −2.62 / −2.45 eV/atom
ΔS_f at 300 / 1000 K	−561 / −516 J K ^{−1} mol ^{−1} per formula unit
S_{vib} (300 K); C_v (1000 K)/3R of Sr ₂ PrSbO ₆	25.0 J K ^{−1} mol ^{−1} atom ^{−1} ; 0.98
Elemental references (PBE, 600 eV)	fcc Sr a = 6.023 Å (expt 6.085); dhcp Pr a = 3.744, c = 12.02 Å (expt 3.673, 11.84); A7 Sb 31.8 Å ³ /atom (expt 30.2); O ₂ d = 1.233 Å, triplet (expt 1.208)

Table S19 — Solar-blind photodetectors used in the benchmark of Fig. 8

Table S19. Solar-blind photodetectors used in the benchmark of Fig. 8 (main text). Responsivity R and specific detectivity D* are quoted at the wavelength λ and bias reported in each source (n/s = not specified in the source); EQE = $R \cdot hc / (q\lambda)$ evaluated at the reported wavelength (250 nm assumed where λ is not specified); “gain” denotes EQE > 100 % (internal photoconductive, phototransistor or avalanche gain), “no gain” EQE ≤ 100 %. Band gaps are those stated in the source or, where absent, standard literature values. This work: E_g = 5.09 eV, R (220 nm, 10 V) = 2.7 mA/W (d = 200 nm), 4.5 mA/W (5 μ m) and 45 mA/W (5 μ m, τ = 10 ns); $D^* = R / (2qJ_{\text{dark}})^{1/2} = 4.8 \times 10^{11}$, 7.9×10^{11} and 7.9×10^{12} Jones with $J_{\text{dark}} = 10^{-10}$ A cm^{−2}.

Device	Year	E_g (eV)	R (A/W)	λ (nm)	Bias (V)	EQE (%)	Class	D* (Jones)	Ref.
diamond: single-crystal CVD diamond	2005	5.50	0.03	n/s	n/s	14.9	no gain	—	S40
β -Ga ₂ O ₃ : sol-gel β -Ga ₂ O ₃ film MSM	2007	4.80	0.1	n/s	n/s	49.6	no gain	—	S39
AlGaIn back-illuminated p-i-n (EQE 89 %)	2013	4.50	0.23	n/s	n/s	114	gain	—	S41
β -Ga ₂ O ₃ : MBE β -Ga ₂ O ₃ MSM, graphene electrodes	2017	4.80	29.8	258	n/s	1.4×10^4	gain	—	S16
Sr ₂ Nb ₃ O ₁₀ : 2D perovskite-oxide nanosheets	2020	3.90	1.2×10^3	270	n/s	5.6×10^5	gain	1.4×10^{14}	S42

a-Ga ₂ O ₃ : annealed amorphous Ga ₂ O ₃ MSM, 32×32 array	2021	4.98	733	254	5	3.6×10^5	gain	3.9×10^{16}	S43
AlGa _{0.4} N: Al _{0.4} Ga _{0.6} N asymmetric-heterostructure MSM, self-powered	2023	4.35	0.04	275	0	18	no gain	3.1×10^{12}	S44
Al _{0.4} YN: p-Gr/i-Al _{0.4} YN/n-Si photovoltaic	2023	4.70	0.013	255	0	6.32	no gain	5.1×10^{12}	S45
ZnGa ₂ O ₄ : NiO/ZnGa ₂ O ₄ heterojunction, self-powered	2024	5.00	0.088	246	0	44.4	no gain	2.0×10^{14}	S46
diamond: single-crystal diamond 8×8 pixel MSM, 13 ns rise	2024	5.47	22.6	n/s	50	1.1×10^4	gain	4.2×10^{14}	S48
β-Ga ₂ O ₃ : suspended-channel β-Ga ₂ O ₃ MSM (0.01 μW cm ⁻²)	2024	4.67	1.5×10^{10}	250	0.1	7.5×10^{12}	gain	6.0×10^{17}	S47
Ga ₂ O ₃ /AlN/AlGa _{0.4} N:Si nBn heterostructure, self-powered, 56 μs	2025	5.02	0.73	245	0	369	gain	2.9×10^{11}	S49
MgZnO/InGaO heterojunction MSM (self-trapped-hole gain)	2025	4.77	1.4×10^4	260	5	6.9×10^6	gain	7.9×10^{17}	S50
Ga ₂ O ₃ : sputtered NiO/Ga ₂ O ₃ heterojunction, self-powered	2026	4.90	0.047	254	0	22.9	no gain	7.5×10^{11}	S52
diamond: all-diamond MSM, boron-doped-diamond transparent electrodes	2026	5.47	0.0498	222	10	27.8	no gain	—	S51

Supplementary References

- (S1) Togo, A.; Chaput, L.; Tada, T.; Tanaka, I. Implementation Strategies in Phonopy and Phono3py. *J. Phys.: Condens. Matter* 2023, 35, 353001. <https://doi.org/10.1088/1361-648X/acd831>.
- (S2) Togo, A.; Chaput, L.; Tanaka, I. Distributions of Phonon Lifetimes in Brillouin Zones. *Phys. Rev. B* 2015, 91, 094306. <https://doi.org/10.1103/PhysRevB.91.094306>.
- (S3) Chaput, L. Direct Solution to the Linearized Phonon Boltzmann Equation. *Phys. Rev. Lett.* 2013, 110, 265506. <https://doi.org/10.1103/PhysRevLett.110.265506>.
- (S4) Kresse, G.; Furthmüller, J. Efficient Iterative Schemes for Ab Initio Total-Energy Calculations Using a Plane-Wave Basis Set. *Phys. Rev. B* 1996, 54, 11169–11186. <https://doi.org/10.1103/PhysRevB.54.11169>.

- (S5) Kresse, G.; Furthmüller, J. Efficiency of Ab-Initio Total Energy Calculations for Metals and Semiconductors Using a Plane-Wave Basis Set. *Comput. Mater. Sci.* 1996, 6, 15–50. [https://doi.org/10.1016/0927-0256\(96\)00008-0](https://doi.org/10.1016/0927-0256(96)00008-0).
- (S6) Blöchl, P. E. Projector Augmented-Wave Method. *Phys. Rev. B* 1994, 50, 17953–17979. <https://doi.org/10.1103/PhysRevB.50.17953>.
- (S7) Kresse, G.; Joubert, D. From Ultrasoft Pseudopotentials to the Projector Augmented-Wave Method. *Phys. Rev. B* 1999, 59, 1758–1775. <https://doi.org/10.1103/PhysRevB.59.1758>.
- (S8) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* 1996, 77, 3865–3868. <https://doi.org/10.1103/PhysRevLett.77.3865>.
- (S9) Madsen, G. K. H.; Singh, D. J. BoltzTraP. A Code for Calculating Band-Structure Dependent Quantities. *Comput. Phys. Commun.* 2006, 175, 67–71. <https://doi.org/10.1016/j.cpc.2006.03.007>.
- (S10) Sanchez, J. E. DEVSIM: A TCAD Semiconductor Device Simulator. *J. Open Source Softw.* 2022, 7 (70), 3898. <https://doi.org/10.21105/joss.03898>.
- (S11) Heyd, J.; Scuseria, G. E.; Ernzerhof, M. Hybrid Functionals Based on a Screened Coulomb Potential. *J. Chem. Phys.* 2003, 118, 8207–8215. <https://doi.org/10.1063/1.1564060>.
- (S12) Krukau, A. V.; Vydrov, O. A.; Izmaylov, A. F.; Scuseria, G. E. Influence of the Exchange Screening Parameter on the Performance of Screened Hybrid Functionals. *J. Chem. Phys.* 2006, 125, 224106. <https://doi.org/10.1063/1.2404663>.
- (S13) Koksai, O.; Tanen, N.; Jena, D.; Xing, H. G.; Rana, F. Measurement of Ultrafast Dynamics of Photoexcited Carriers in β -Ga₂O₃ by Two-Color Optical Pump-Probe Spectroscopy. *Appl. Phys. Lett.* **2018**, 113, 252102. <https://doi.org/10.1063/1.5058164>.
- (S14) Kobayashi, K.-I.; Kimura, T.; Sawada, H.; Terakura, K.; Tokura, Y. Room-Temperature Magnetoresistance in an Oxide Material with an Ordered Double-Perovskite Structure. *Nature* 1998, 395, 677–680. <https://doi.org/10.1038/27167>.
- (S15) Michaelson, H. B. The Work Function of the Elements and Its Periodicity. *J. Appl. Phys.* 1977, 48, 4729–4733. <https://doi.org/10.1063/1.323539>.
- (S16) Singh Pratiyush, A.; Krishnamoorthy, S.; Vishnu Solanke, S.; Xia, Z.; Muralidharan, R.; Rajan, S.; Nath, D. N. High Responsivity in Molecular Beam Epitaxy Grown β -Ga₂O₃ Metal Semiconductor Metal Solar Blind Deep-UV Photodetector. *Appl. Phys. Lett.* 2017, 110, 221107. <https://doi.org/10.1063/1.4984904>.
- (S17) Zheng, J.; Lin, C.; Lin, C.; Hautier, G.; Guo, R.; Huang, B. Unravelling Ultralow Thermal Conductivity in Perovskite Cs₂AgBiBr₆: Dominant Wave-like Phonon Tunnelling and Strong Anharmonicity. *npj Comput. Mater.* 2024, 10, 30. <https://doi.org/10.1038/s41524-024-01211-y>.
- (S18) Fergus, J. W. Oxide Materials for High Temperature Thermoelectric Energy Conversion. *J. Eur. Ceram. Soc.* 2012, 32, 525–540. <https://doi.org/10.1016/j.jeurceramsoc.2011.10.007>.
- (S19) Wu, H.; Shi, X.-L.; Liu, W.-D.; Li, M.; Gao, H.; Zhou, W.; Shao, Z.; Wang, Y.; Liu, Q.; Chen, Z.-G. Double Perovskite Pr₂CoFeO₆ Thermoelectric Oxide: Roles of Sr-Doping and Micro/Nanostructuring. *Chem. Eng. J.* 2021, 425, 130668. <https://doi.org/10.1016/j.cej.2021.130668>.
- (S20) Wang, V.; Xu, N.; Liu, J.-C.; Tang, G.; Geng, W.-T. VASPKIT: A User-Friendly Interface Facilitating High-Throughput Computing and Analysis Using VASP Code. *Comput. Phys. Commun.* 2021, 267, 108033. <https://doi.org/10.1016/j.cpc.2021.108033>.

- (S21) Pugh, S. F. XCII. Relations between the Elastic Moduli and the Plastic Properties of Polycrystalline Pure Metals. *Lond. Edinb. Dublin Philos. Mag. J. Sci.* 1954, 45 (367), 823–843. <https://doi.org/10.1080/14786440808520496>.
- (S22) Chen, X.-Q.; Niu, H.; Li, D.; Li, Y. Modeling Hardness of Polycrystalline Materials and Bulk Metallic Glasses. *Intermetallics* 2011, 19 (9), 1275–1281. <https://doi.org/10.1016/j.intermet.2011.03.026>.
- (S23) Xu, B.; Tian, Y. Ultrahardness: Measurement and Enhancement. *J. Phys. Chem. C* 2015, 119 (10), 5633–5638. <https://doi.org/10.1021/acs.jpcc.5b00017>.
- (S24) Zener, C. *Elasticity and Anelasticity of Metals*; The University of Chicago Press: Chicago, IL, 1948.
- (S25) Tersoff, J. Schottky Barrier Heights and the Continuum of Gap States. *Phys. Rev. Lett.* 1984, 52, 465–468. <https://doi.org/10.1103/PhysRevLett.52.465>
- (S26) Schleife, A.; Fuchs, F.; Rödl, C.; Furthmüller, J.; Bechstedt, F. Branch-Point Energies and Band Discontinuities of III-Nitrides and III-/II-Oxides from Quasiparticle Band-Structure Calculations. *Appl. Phys. Lett.* 2009, 94, 012104. <https://doi.org/10.1063/1.3059569>
- (S27) Walukiewicz, W. Intrinsic Limitations to the Doping of Wide-Gap Semiconductors. *Physica B* 2001, 302–303, 123–134. [https://doi.org/10.1016/S0921-4526\(01\)00417-3](https://doi.org/10.1016/S0921-4526(01)00417-3)
- (S28) Varley, J. B.; Janotti, A.; Franchini, C.; Van de Walle, C. G. Role of Self-Trapping in Luminescence and p-Type Conductivity of Wide-Band-Gap Oxides. *Phys. Rev. B* 2012, 85, 081109(R). <https://doi.org/10.1103/PhysRevB.85.081109>
- (S29) Wang, L.; Maxisch, T.; Ceder, G. Oxidation Energies of Transition Metal Oxides within the GGA+U Framework. *Phys. Rev. B* 2006, 73, 195107. <https://doi.org/10.1103/PhysRevB.73.195107>
- (S30) Wang, A.; Kingsbury, R.; McDermott, M.; Horton, M.; Jain, A.; Ong, S. P.; Dwaraknath, S.; Persson, K. A. A Framework for Quantifying Uncertainty in DFT Energy Corrections. *Sci. Rep.* 2021, 11, 15496. <https://doi.org/10.1038/s41598-021-94550-5>
- (S31) Chase, M. W., Jr. *NIST-JANAF Thermochemical Tables*, 4th ed.; J. Phys. Chem. Ref. Data, Monograph 9; American Institute of Physics: Woodbury, NY, 1998.
- (S32) Haynes, W. M., Ed. *CRC Handbook of Chemistry and Physics*, 97th ed.; CRC Press: Boca Raton, FL, 2016.
- (S33) Kittel, C. *Introduction to Solid State Physics*, 8th ed.; Wiley: Hoboken, NJ, 2005.
- (S34) Gschneidner, K. A., Jr. *Physical Properties and Interrelationships of Metallic and Semimetallic Elements*. *Solid State Phys.* 1964, 16, 275–426. [https://doi.org/10.1016/S0081-1947\(08\)60518-4](https://doi.org/10.1016/S0081-1947(08)60518-4)
- (S35) Huber, K. P.; Herzberg, G. *Molecular Spectra and Molecular Structure IV. Constants of Diatomic Molecules*; Van Nostrand Reinhold: New York, 1979.
- (S36) Wong-Ng, W.; Kaduk, J. A.; Luong, M.; Huang, Q. X-Ray Diffraction Study and Powder Patterns of Double-Perovskites Sr_2RSbO_6 (R = Pr, Nd, Sm, Eu, Gd, Dy, Ho, Y, Er, Tm, Yb, and Lu). *Powder Diffr.* 2014, 29, 371–378. <https://doi.org/10.1017/S0885715614000566>
- (S37) Faik, A.; Iturbe-Zabalzo, E.; Urcelay, I.; Igartua, J. M. Crystal Structures and High-Temperature Phase Transitions of the New Ordered Double Perovskites $\text{Sr}_2\text{SmSbO}_6$ and $\text{Sr}_2\text{LaSbO}_6$. *J. Solid State Chem.* 2009, 182, 2656–2663. <https://doi.org/10.1016/j.jssc.2009.07.021>
- (S38) Bharti, C.; Sinha, T. P. Dielectric Properties of Rare Earth Double Perovskite Oxide $\text{Sr}_2\text{CeSbO}_6$. *Solid State Sci.* 2010, 12, 498–502. <https://doi.org/10.1016/j.solidstatesciences.2009.12.014>
- (S39) Kokubun, Y.; Miura, K.; Endo, F.; Nakagomi, S. Sol-Gel Prepared $\beta\text{-Ga}_2\text{O}_3$ Thin Films for Ultraviolet Photodetectors. *Appl. Phys. Lett.* 2007, 90, 031912. <https://doi.org/10.1063/1.2432946>

- (S40) Balducci, A.; Marinelli, M.; Milani, E.; Morgada, M. E.; Tucciarone, A.; Verona-Rinati, G.; Angelone, M.; Pillon, M. Extreme Ultraviolet Single-Crystal Diamond Detectors by Chemical Vapor Deposition. *Appl. Phys. Lett.* 2005, 86, 193509. <https://doi.org/10.1063/1.1927709>
- (S41) Cicek, E.; McClintock, R.; Cho, C. Y.; Rahnema, B.; Razeghi, M. $\text{Al}_x\text{Ga}_{1-x}\text{N}$ -Based Back-Illuminated Solar-Blind Photodetectors with External Quantum Efficiency of 89%. *Appl. Phys. Lett.* 2013, 103, 191108. <https://doi.org/10.1063/1.4829065>
- (S42) Li, S.; Zhang, Y.; Yang, W.; Liu, H.; Fang, X. 2D Perovskite $\text{Sr}_2\text{Nb}_3\text{O}_{10}$ for High-Performance UV Photodetectors. *Adv. Mater.* 2020, 32, 1905443. <https://doi.org/10.1002/adma.201905443>
- (S43) Qin, Y.; Li, L.-H.; Yu, Z.; Wu, F.; Dong, D.; Guo, W.; Zhang, Z.; Yuan, J.-H.; Xue, K.-H.; Miao, X.; Long, S. Ultra-High Performance Amorphous Ga_2O_3 Photodetector Arrays for Solar-Blind Imaging. *Adv. Sci.* 2021, 8, 2101106. <https://doi.org/10.1002/adv.202101106>
- (S44) Gu, Y.; Guo, J.; Ye, B.; Yang, X.; Xie, F.; Qian, W.; Zhang, X.; Lu, N.; Liu, Y.; Yang, G. Self-Powered AlGaIn -Based MSM Solar-Blind Ultraviolet Photodetectors with High Al-Content $\text{Al}_x\text{Ga}_{1-x}\text{N}/\text{Al}_y\text{Ga}_{1-y}\text{N}$ Asymmetrical Heterostructure. *Appl. Phys. Lett.* 2023, 123, 232103. <https://doi.org/10.1063/5.0170741>
- (S45) Huang, L.; Huang, S.; Lin, J.; Wang, Z.; Chen, Z.; Zheng, W.; Wei, X.; Hu, Q. Band Gap Engineering of $\text{Al}_x\text{In}_y\text{N}$ Films for Solar-Blind Ultraviolet Photodetection. *ACS Appl. Electron. Mater.* 2023, 5, 1106–1113. <https://doi.org/10.1021/acsaem.2c01581>
- (S46) Khan, T.; Arora, K.; Agarwal, R.; Muduli, P. K.; Chu, Y.-H.; Horng, R. H.; Singh, R. Unveiling Superior Solar-Blind Photodetection with a $\text{NiO}/\text{ZnGa}_2\text{O}_4$ Heterojunction Diode. *ACS Appl. Mater. Interfaces* 2024, 16, 57290–57301. <https://doi.org/10.1021/acsaem.2c01581>
- (S47) Li, X. X.; Wu, Z.; Fang, Y.; Huang, S.; Fang, C.; Wang, Y.; Zeng, X.; Yang, Y.; Hao, Y.; Liu, Y.; Han, G. Ga_2O_3 Solar-Blind Deep-Ultraviolet Photodetectors with a Suspended Structure for High Responsivity and High-Speed Applications. *Research* 2024, 7, 0546. <https://doi.org/10.34133/research.0546>
- (S48) Qiu, M.; Jia, Z.; Yang, M.; Li, M.; Shen, Y.; Liu, C.; Nishimura, K.; Jiang, N.; Wang, B.; Lin, C.-T.; Yuan, Q. High-Performance Single Crystal Diamond Pixel Photodetector with Nanosecond Rise Time for Solar-Blind Imaging. *Diamond Relat. Mater.* 2024, 144, 110996. <https://doi.org/10.1016/j.diamond.2024.110996>
- (S49) Wang, H. B.; Zhou, C.; Li, P.; Yang, L.; Ma, J.; Akaike, R.; Miyake, H.; Xu, H.; Liu, Y. High-Sensitivity and Fast-Response Solar-Blind Photodetectors via Band Offset Engineering for Motion Tracking. *Nat. Commun.* 2025, 16, 8175. <https://doi.org/10.1038/s41467-025-63683-w>
- (S50) Bi, Z.; Liu, Z.; Chen, Z.; Chen, Y.; Zhou, J.; Ruan, S. High-Performance MSM Photodetector Based on $\text{MgZnO}/\text{InGaO}$ Heterojunction for Solar-Blind UV Detection. *Appl. Phys. Lett.* 2025, 127, 163306. <https://doi.org/10.1063/5.0289829>
- (S51) Gao, Y.; Tang, Z.; Liu, Z.; Jiang, F.; Zhang, P.; Li, Y.; Chen, Z.; Liu, T.; Wang, X.; Ao, J. All-Diamond Transparent Solar-Blind UV Photodetector with Thin Boron-Doped Diamond Film as Electrodes. *Opt. Lett.* 2026, 51, 2628–2631. <https://doi.org/10.1364/OL.596496>
- (S52) Liu, L.; Shen, K.; Su, H.; Xu, J.; Lin, J.; Li, Y.; Zhang, S.; Lan, L.; Peng, J. $\text{NiO}/\text{Ga}_2\text{O}_3$ Heterojunction with Tunable Oxygen Vacancies for Efficient Self-Powered Solar-Blind UV Detection. *Materials* 2026, 19, 530. <https://doi.org/10.3390/ma19030530>